# Vaibhav Balloli

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## Research Focus

AI researcher focusing on reliability, safety, experience, and usability of AI systems in human-AI in social good applications. My research has always been deployed to alleviate specific societal challenges. My current work focuses on improving conversational (human-LLM) and agentic capabilities for maternal health.

**Research Topics:** LLMs & {Evals, Alignment, Safety, and Deployment}, human-AI interaction

## Education

**University of Michigan, Ann Arbor** *2023 - Present*
*Ph.D. Candidate, M.S. in Computer Science and Engineering*
Advisor: Prof. Elizabeth Bondi-Kelly
Courses: Reinforcement Learning Theory, Causality and ML, NLP, HCI, AI, Advanced Data Mining

**BITS Pilani, Hyderabad Campus** *2016 - 2020*
*B.E in Electronics and Communication Engineering*

## Research Experience

**Microsoft Research Redmond, AI Frontiers** *May 2025 - Aug 2025*
*Research Intern*
**Advisors:** Hussein Mozannar, Adam Fourney, Gagan Bansal, Saleema Amershi & Eric Horvitz
**Topic:** Multi-turn Reinforcement Learning with Large Language Models Agents, Agent evaluations

**Microsoft Research India** *June 2022 - June 2023*
*Research Fellow*
**Advisors:** Dr. Akshay Nambi & Tanuja Ganu
**Topics:** Reinforcement Learning, Large Language Models.
**Projects and Press:** Vasudha, VeLLM: VeLLM Press - Satya Nadella, Times of India
**Impact:** VeLLM is currently being deployed in schools across India

**Microsoft Research India** *June 2021 - June 2022*
*Societal impact through Cloud and AI Fellow*
**Advisors:** Dr. Akshay Nambi, Tanuja Ganu & Dr. Venkat Padmanabhan
**Topics:** Computer Vision, Bandits, Localization
**Projects and Press:** HAMS: HAMS Press - Punjab News Express
**Impact:** HAMS is successfully deployed across India - Dashboard (300,000 users)

## Working papers

1. **Towards Improving Interactivity in Multi-Turn Human-Agent Systems**
   **Vaibhav Balloli**, Hussein Mozannar, Adam Fourney, Gagan Bansal, Saleema Amershi, Eric Horvitz
   *Topics: Large Language Models, Agents, GRPO*
2. **RELIANCE: Curating and Evaluating Reproductive Health Information on Social Media**
   **Vaibhav Balloli**, Laura Peyton Ellis, Vishala Mishra, Alice Chi, Alex Friedman Peahl, Elizabeth Bondi-Kelly
   *Topics: Large Language Models, Reproductive Health, Multimodal eval.*
3. **"Where is this coming from?" Uncovering Trustworthiness Ideals in AI-powered Peripartum Information Seeking**
   **Vaibhav Balloli**, Julia Erickson, Xinyi Li, Erin van Liemt, Alex Friedman Peahl, Elizabeth Bondi-Kelly
   *Topics: Focus groups, Qualitative Analysis, human-AI system design*

## Selected Publications

4. **PATHFinder Agent for Tailored Prenatal Care**
   **Vaibhav Balloli**, Carissa Samuel, Samia Abdelnabi, Alex Peahl, Elizabeth Bondi-Kelly.
   *Topics: Large Language Models Agents, Long-context optimization, Deployment*
   ***Interactive Health, 2026***

3. **Bridging the Gap: Dynamic Learning Strategies for Improving Multilingual Performance in LLMs**
   Somnath Kumar*, **Vaibhav Balloli***, Mercy Ranjit, Kabir Ahuja, Tanuja Ganu, Sunayana Sitaram, Kalika Bali, Akshay Nambi
   *Topics: Large Language Models, Multilingual Evaluation, Human Feedback, Representation Learning*
   ***Oral presentation, COLING 2025***

2. **Are They the Same Picture? Adapting Concept Bottleneck Models for Human-AI Collaboration in Image Retrieval**
   **Vaibhav Balloli**, Sara Beery, Elizabeth Bondi-Kelly
   *Topics: human-AI Collaboration, Representation Learning, Interpretability, Wildlife Conservation*
   ***IJCAI'24 Human-Centered AI Track, CV4Animals Workshop@CVPR'24*** | **Code** | **Project**
   **[<5% Acceptance Rate]**

1. **Chanakya: Learning Tradeoffs for Adaptive Streaming Perception via Contextual Bandits**
   Anurag Ghosh, **Vaibhav Balloli**, Akshay Nambi, Aditya Singh, Tanuja Ganu
   *Topics: Object detection and tracking, Computer Vision, Multi-arm bandits*
   ***NeurIPS'23*** | **Talk** | **Code**

## Select Awards and Honors

- Exceptional Preliminary exam (advancement to candidacy exam) *2025*
- Presented poster at Society-centered AI workshop at Google, Mountain View *2025*
- 2nd Place in the Google x MHacks Hackathon - $1500 Press: Mentioned by **labs.google** *2024*
- **Press:** VeLLM mentioned by Satya Nadella and featured in Times Of India, FirstPost *2024*
- Selected for the Harvard/MIT - HAIST/MAIA Intro Fellowship on AI Safety *2023*
- **Press:** HAMS Automated License Testing featured in Punjab News Express *2022*
- **Winner** of Microsoft Global Hackathon, 2021(Future Of Edge Computing Track) *2021*
- Co-founded **SmartCampus** which successfully managed **₹25 million** in transactions *2020*

## Service

- Conference Reviewer: NeurIPS'25, IJCAI'25, ICML'25, AAMAS'25, ICLR'25 and NeurIPS'24.
- Workshop Reviewer: CV4Animals@CVPR'24, CompSust@ NeurIPS'23.

## Software

1. autogen-openaiext-client Autogen community project for working with external LLMs through OpenAI API.

2. **Automated License Testing - Microsoft Research India**.
   *Microsoft Research India*
   Visual SLAM, Object Detection, and Trajectory analysis.